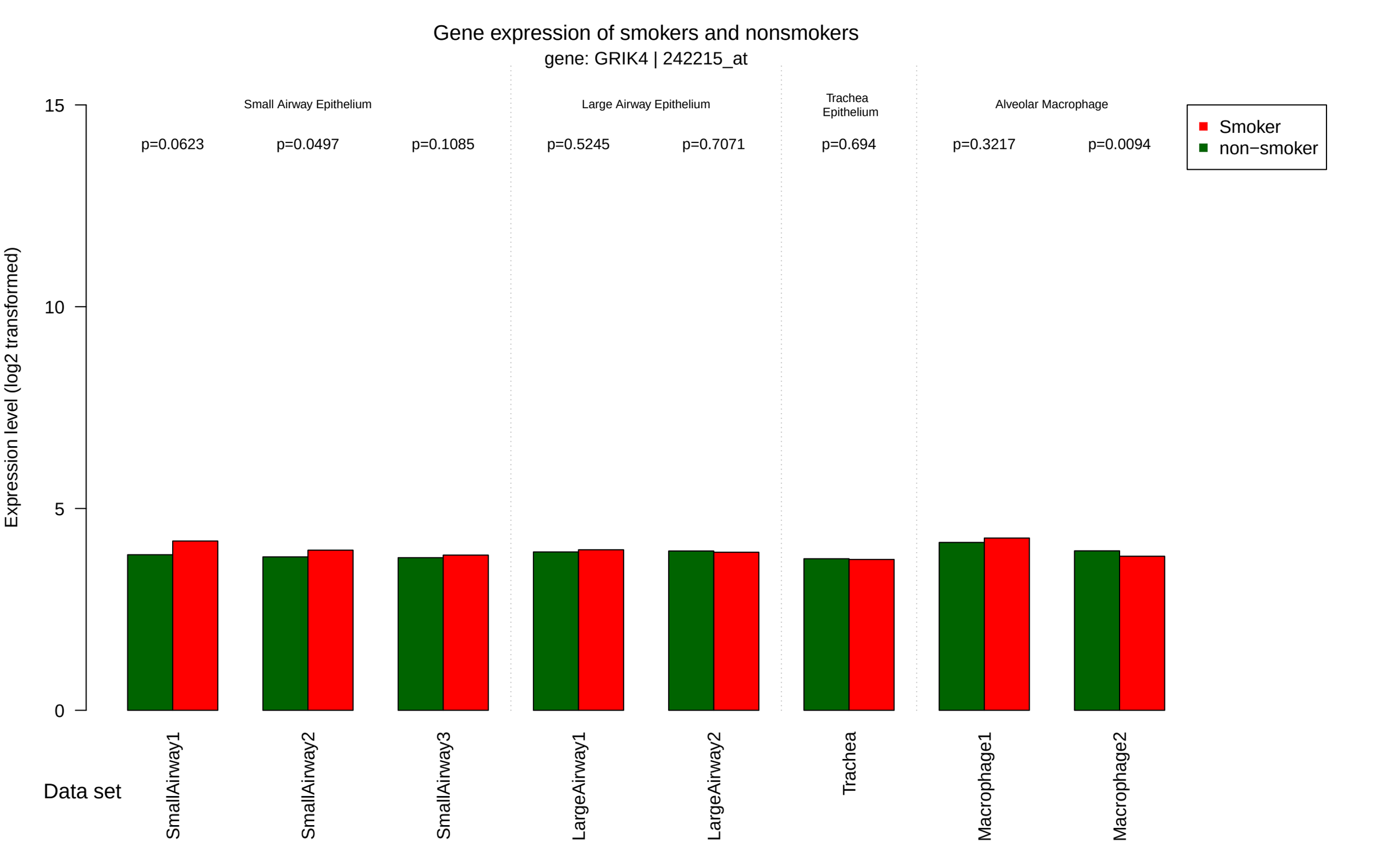
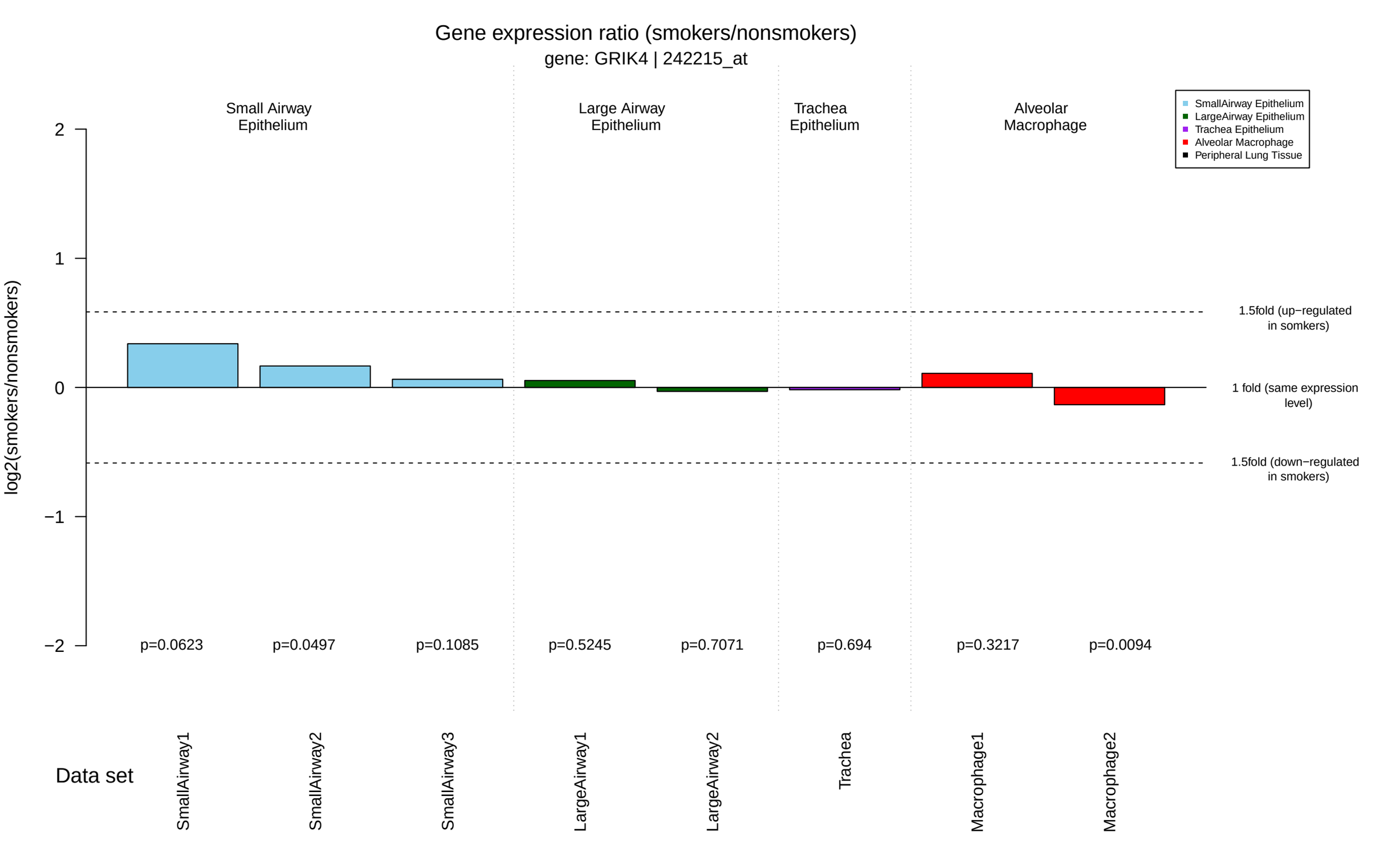
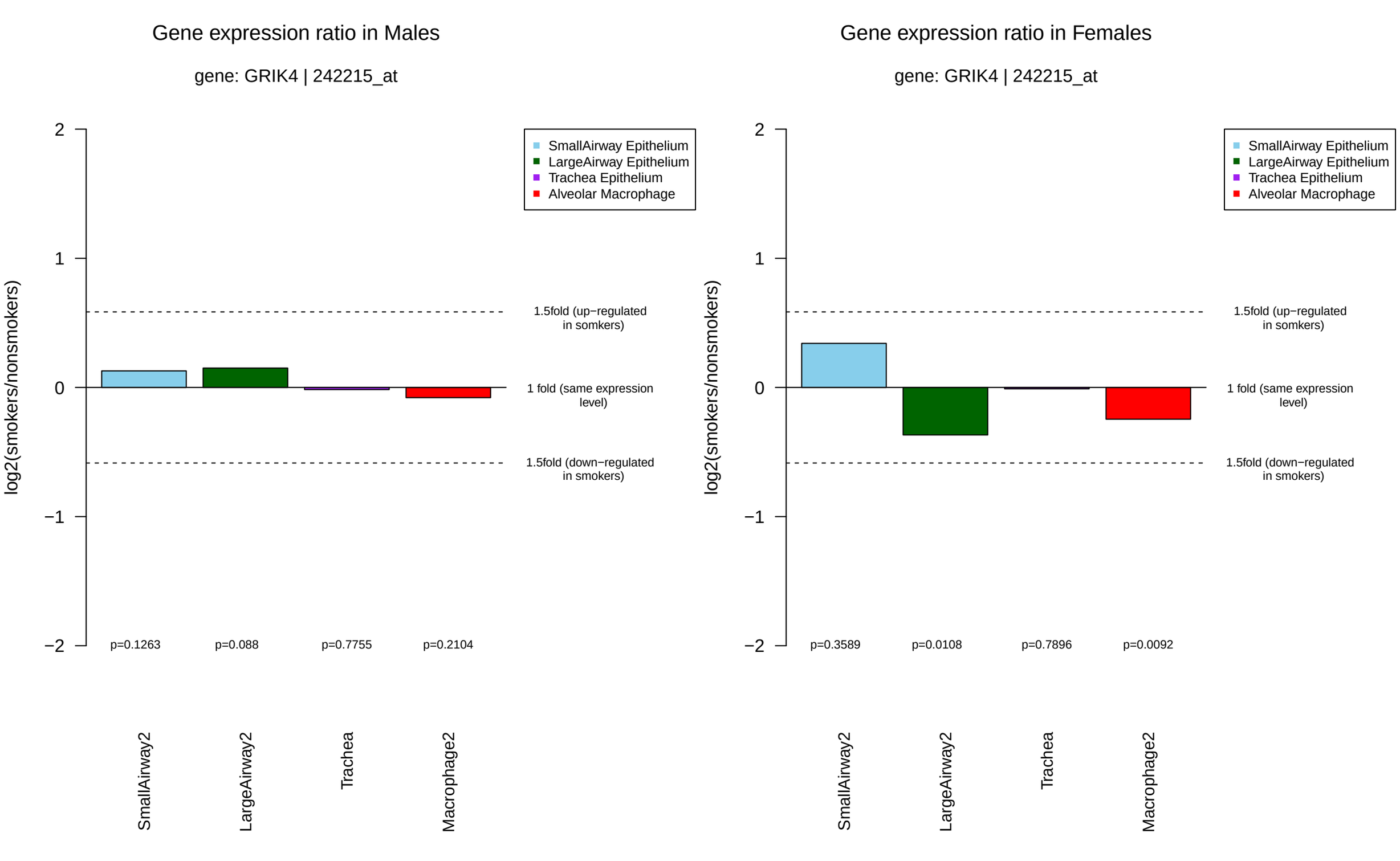
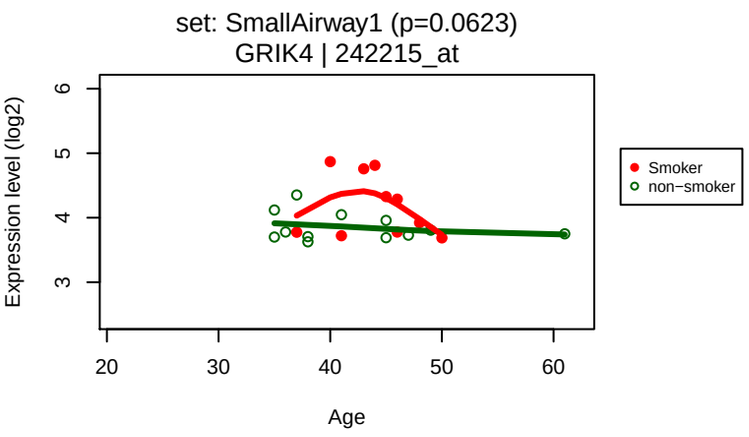
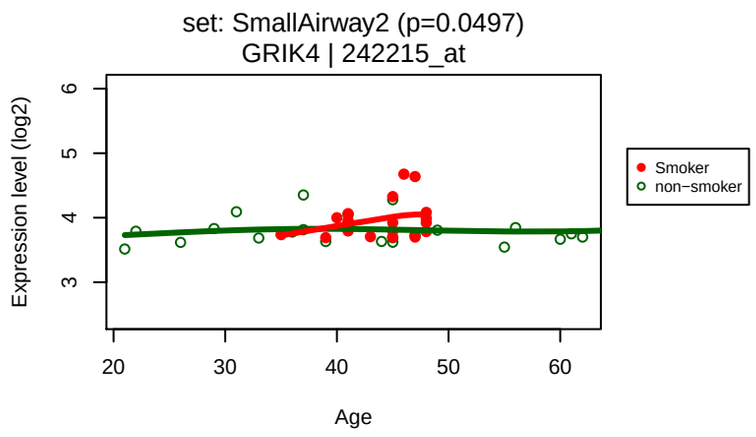
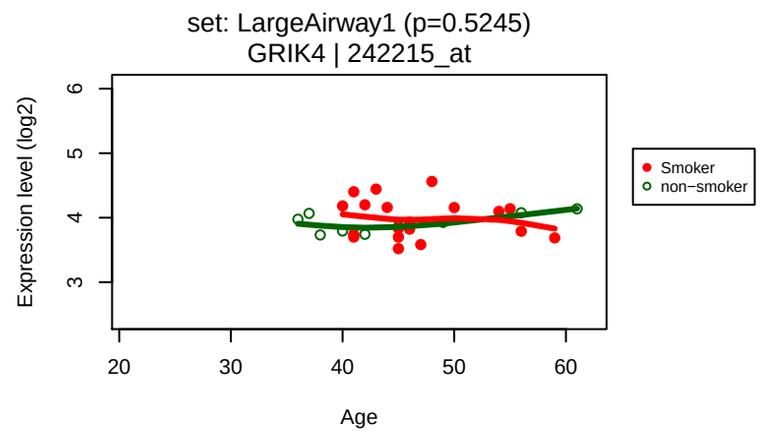

Gene expression of smokers and nonsmokers
gene: GRIK4 | 242215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0623
p=0.0497
p=0.1085
p=0.5245
p=0.7071
p=0.694
p=0.3217
p=0.0094
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

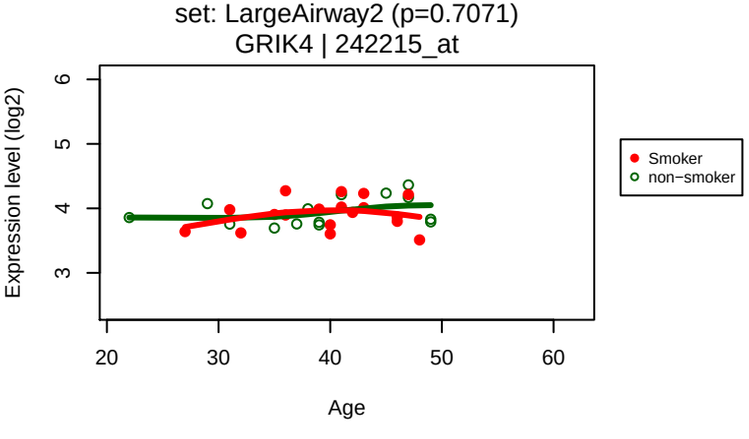

Gene expression ratio (smokers/nonsmokers)
gene: GRIK4 | 242215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0623
p=0.0497
p=0.1085
p=0.5245
p=0.7071
p=0.694
p=0.3217
p=0.0094
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

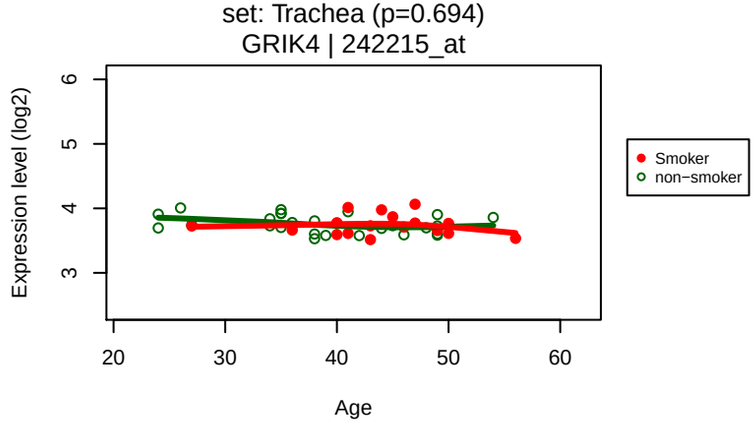

Gene expression ratio in Males
gene: GRIK4 | 242215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1263
p=0.088
p=0.7755
p=0.2104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIK4 | 242215_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3589
p=0.0108
p=0.7896
p=0.0092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

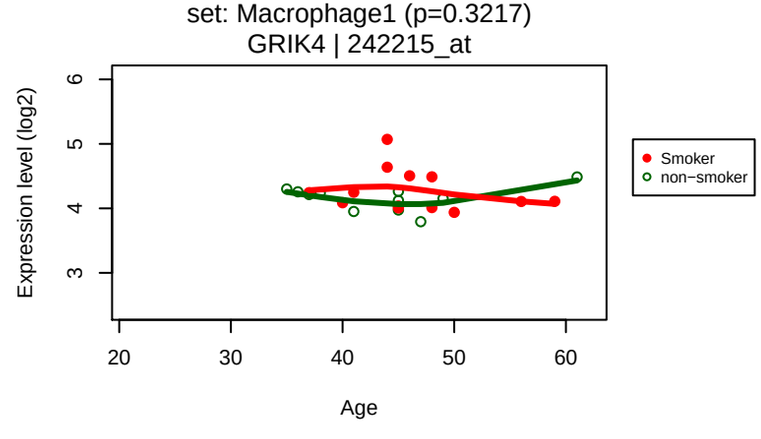

set: SmallAirway1 (p=0.0623)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker

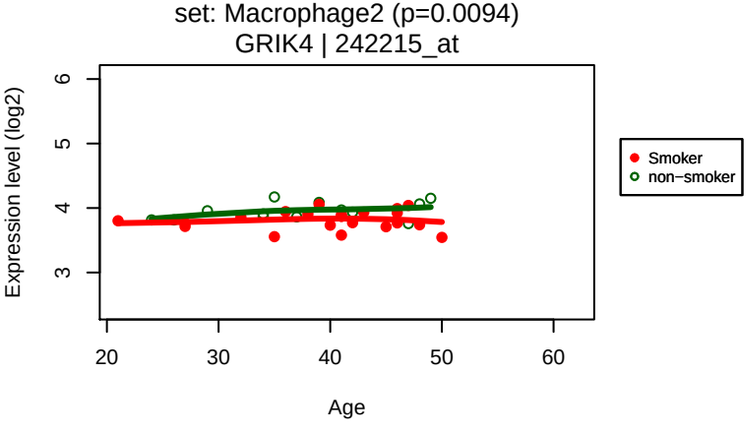

set: SmallAirway2 (p=0.0497)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker

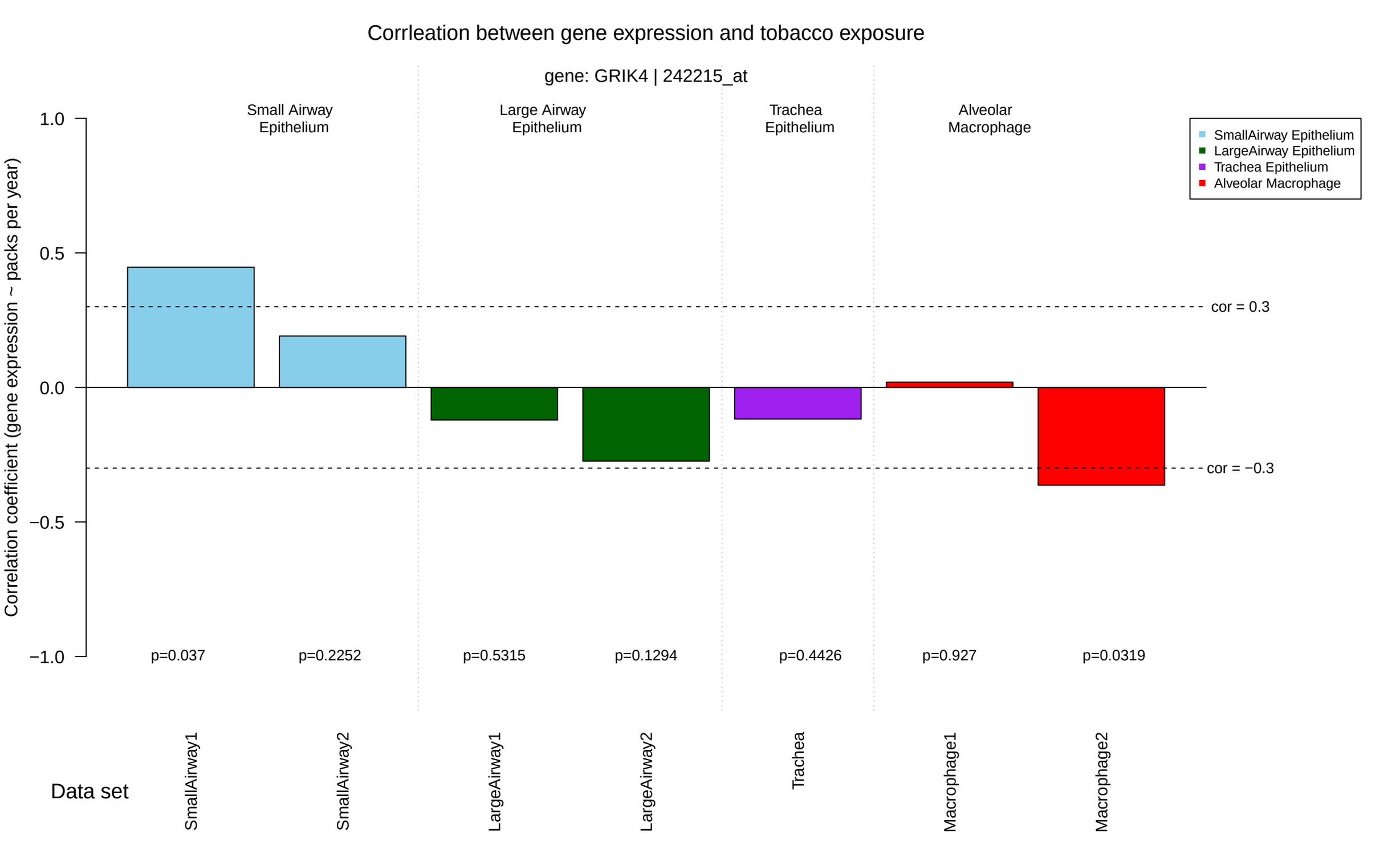

set: LargeAirway1 (p=0.5245)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7071)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.694)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3217)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0094)
GRIK4 | 242215_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIK4 | 242215_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.037
p=0.2252
p=0.5315
p=0.1294
p=0.4426
p=0.927
p=0.0319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2